Amartya (Marty) Mukherjee

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Research Statement

Applied Mathematics PhD candidate and machine learning researcher specializing in diffusion models, time-series generative modeling, stochastic control, and outlier generation. Research experience at RBC Borealis and Huawei, with publications at ICML, TMLR, and CDC, and a U.S. patent. Seeking research scientist opportunities beginning in September 2027.

Education

University of Waterloo **September 2022 – August 2027 (Expected)**
Doctor of Philosophy, Applied Mathematics, Supervisor: Jun Liu

University of Waterloo **September 2017 – April 2022**
Bachelor of Mathematics, Applied Mathematics with Scientific Computing, Minor in Statistics

Preprints

- **Amartya Mukherjee**, Maxwell Fitzsimmons, David C. Del Rey Fernández, and Jun Liu (2026). *Rigorous Error Certification for Neural PDE Solvers: From Empirical Residuals to Solution Guarantees.* arXiv:2603.19165
- Shuxiu Jia, **Amartya Mukherjee**, Yating Yuan, and Jun Liu (2026). *Volumetric Harmonic Field Navigation for Quadrotors.* arXiv:2609.15680

Publications

- **Amartya Mukherjee**, Tristan Milne, Kry Yik Chau Lui, Stephanie Hazlewood, and Jun Liu (2026). *Score-Based Outlier Generation via Controlling the Radon-Nikodym Derivative.* IEEE Conference on Decision and Control (CDC 2026). [Link]
- **Amartya Mukherjee**, Maxwell Fitzsimmons, David C. Del Rey Fernández, and Jun Liu (2026). *Learning Lyapunov Operators for Nonlinear Systems.* IEEE Conference on Decision and Control (CDC 2026). [Link]
- **Amartya Mukherjee** and Jun Liu (2026). *Almost Sure Convergence Analysis of Stochastic Gradient Methods with Clipping and Additive Noise.* IEEE Conference on Decision and Control (CDC 2026). [Link]
- **Amartya Mukherjee**, Ruizhi Deng, He Zhao, Yuzhen Mao, Leonid Sigal, and Frederick Tung (2026). *ADiff4TPP: Asynchronous Diffusion Models for Temporal Point Processes.* Transactions on Machine Learning Research. [OpenReview]
- **Amartya Mukherjee**, Thanin Quartz, and Jun Liu (2025). *Manifold-Guided Stabilization of Nonlinear Dynamical Systems with Diffusion Models.* American Control Conference (ACC 2025). [Link]
- Yiming Meng, Ruikun Zhou, **Amartya Mukherjee**, Maxwell Fitzsimmons, Christopher Song, and Jun Liu (2024). *Physics-Informed Neural Policy Iteration: Algorithms, Convergence, and Verification.* 41st International Conference on Machine Learning (ICML 2024). [Link]
- **Amartya Mukherjee** and Jun Liu (2023). *Actor-Critic Methods using Physics-Informed Neural Networks: Control of a 1D PDE Model for Fluid-Cooled Battery Packs.* ICML Workshop on New Frontiers in Learning, Control, and Dynamical Systems. [OpenReview]
- **Amartya Mukherjee** and Jun Liu (2023). *Bridging Physics-Informed Neural Networks with Reinforcement Learning: Hamilton-Jacobi-Bellman Proximal Policy Optimization (HJBPPO).* ICML Workshop on New Frontiers in Learning, Control, and Dynamical Systems. [OpenReview]
- **Amartya Mukherjee**, Yusuf Aydogdu, Thambirajah Ravichandran, and Navaratnam Sri Namachchivaya (2022). *Stochastic Parameterization Using Compressed Sensing: Application to the Lorenz-96 Atmospheric Model.* Tellus A: Dynamic Meteorology and Oceanography, 74(2022), pp.300–317. DOI: http://doi.org/10.16993/tellusa.42

Patents

- Sheral Kumar, **Amartya Mukherjee**, Seel Patel, Rui Xiang Chai, Wentao Liu, Yuanhao Yu, Yang Wang, Jin Tang. *Methods and Devices for Extracting Motion Vector Data from Compressed Video Data.* US Patent 11,729,395 B2, filed November 26, 2021, published August 15, 2023. [Google Patents]

Research Internship Experience

MITACS Research Intern, RBC Borealis **Feb 2026– Sep 2026**

Mentor: Drs. Stephanie Hazlewood, Tristan Milne, Kry Yik-Chau Lui

- Developed a likelihood-space control method for diffusion models that generates rare and tail-event scenarios without retraining or finetuning the underlying model.
- Designed and evaluated diffusion models for financial time-series outlier synthesis, reducing tail-distribution error by 94% and marginal distribution error by 77% relative to the strongest GAN baselines on S&P 500 sector benchmarks.
- Derived theoretical guarantees for the controlled sampling procedure; Research published in IEEE CDC 2026.

Machine Learning Research Intern, RBC Borealis **Sep 2024– Jan 2025**

Mentor: Drs. Frederick Tung, Ruizhi Deng, He Zhao

- Developed ADiff4TPP, a diffusion-transformer framework for forecasting temporal point processes under arbitrary observed and predicted event splits.
- Designed an asynchronous noise schedule and transformer-based latent representation for long-horizon event forecasting, outperforming nine baselines by 14% across five datasets.
- Formulated partial differential equations linking the diffusion process to the stochastic prediction of future events given past observations; Research published in Transactions on Machine Learning Research.

Computer Vision Researcher, Noah's Ark Lab at Huawei Technologies Canada **May 2021– Dec 2021**

Mentor: Drs. Yuanhao Yu, Wentao Liu

- Implemented solutions for accelerating video AI tasks on smartphones by focusing on motion vector extraction.
- Integrated customized H.264 and HEVC video decoders into the Android Media framework, enabling the extraction of motion vector information from video streams.
- Achieved significant performance improvements by reducing computation times for optical flow and action recognition models by 55% and 34%, respectively, through the efficient use of motion vector data; US Patent Published.

Research Intern, Fields CQAM Laboratory for Inference and Prediction **May 2020– Apr 2021**

Mentor: Professor N. Sri Namachchivaya

- Developed a data-driven stochastic parameterization framework for multi-scale atmospheric models to represent unresolved small-scale dynamics in terms of resolved variables.
- Integrated sparse optimization and auto-regressive noise modeling, demonstrating that compressed sensing with stochastic residual modeling outperforms traditional polynomial regression in trajectory forecasting and distribution matching.
- Reduced the mean squared prediction error of Rossby waves by 21% using Compressed Sensing and further by 64% using Ensemble Kalman Filter; Research Published in Tellus A: Dynamic Meteorology and Oceanography.

Teaching Experience

MATH 237 - Calculus 3 for Honors Mathematics **May 2025– Aug 2025**

Course Instructor and Coordinator - Faculty of Mathematics

- Led and coordinated an undergraduate multivariable calculus course, overseeing instruction, assessments, and overall course delivery.
- Designed assignments and examinations incorporating applications from statistics, convex optimization, and partial differential equations to connect core mathematical concepts with more specialized and applicable problems.

AMATH 495/900 - Mathematics of Deep Learning **Jan 2025– Apr 2025**

Teaching Assistant - Department of Applied Mathematics

- Evaluated and provided technical feedback on graduate and senior undergraduate projects in deep learning theory, including convergence of SGD and accelerated methods, diffusion models, Neural ODEs, and generalization theory.
- Developed assignment solutions and assessed student work on approximation theory, convergence guarantees for gradient-based optimization, and statistical learning theory, including VC dimension and Rademacher complexity.

Honors and scholarships

- Physical AI Graduate Scholarship - 15,000 CAD
- MITACS Accelerate Grant - 30,000 CAD
- NeurIPS 2025 Top Reviewer - 500 USD
- American Control Conference 2025 Student Travel Grant - 650 USD
- International Conference on Machine Learning 2023 Student Travel Grant - 1,030 USD
- President's Scholarship (University of Waterloo) - 2,000 CAD
- Founder's Scholarship (K. International School Tokyo) - 300,000 JPY